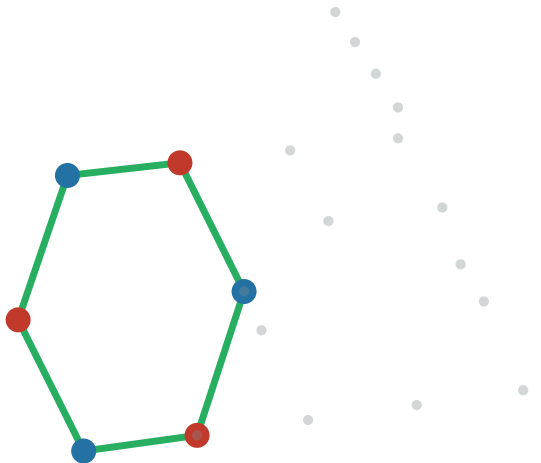
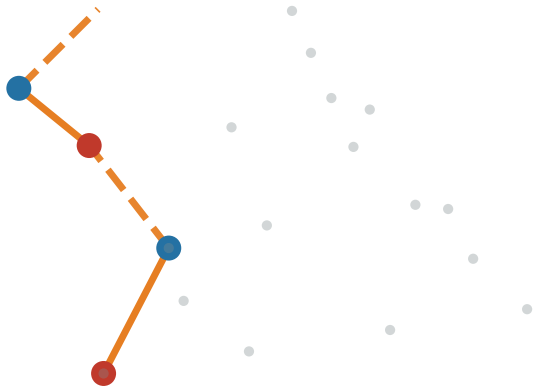


Both loops flip alternating spins and preserve the ice rule — only their torus transport differs

Contractible hexagon (physics)



Winding four-loop (artifact)



raised–lowered balance: $\delta = \rho + \text{NL} = 0$

offset by half a cell: $\delta = (0, 0, -\frac{1}{2}) \neq 0$

● raised spin ● lowered spin - - - bond wrapping the torus